

ALGORITHMIC & PORTFOLIO TRADING

Mauricio Labadie, PhD
Algorithmic Quant/Trader
Electronic Broker (Investment Bank)
London, UK

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1. Introduction

Market Order (MOs)

- Instruction to buy (sell) a number of shares at any current price available in the market
- It ensures complete execution
- But consumes liquidity
 - MOs “pay the spread”
- The final execution price could be very different to the current market price



deal-too.com

Limit Orders (LOs)

- Instruction to buy (sell) a number of shares at a specified price or less (more)
- It includes a limit price e.g. 82 USD
- It provides liquidity to MOs
 - LOs “earn/capture the spread”
- But there is a risk of incomplete (or no) execution



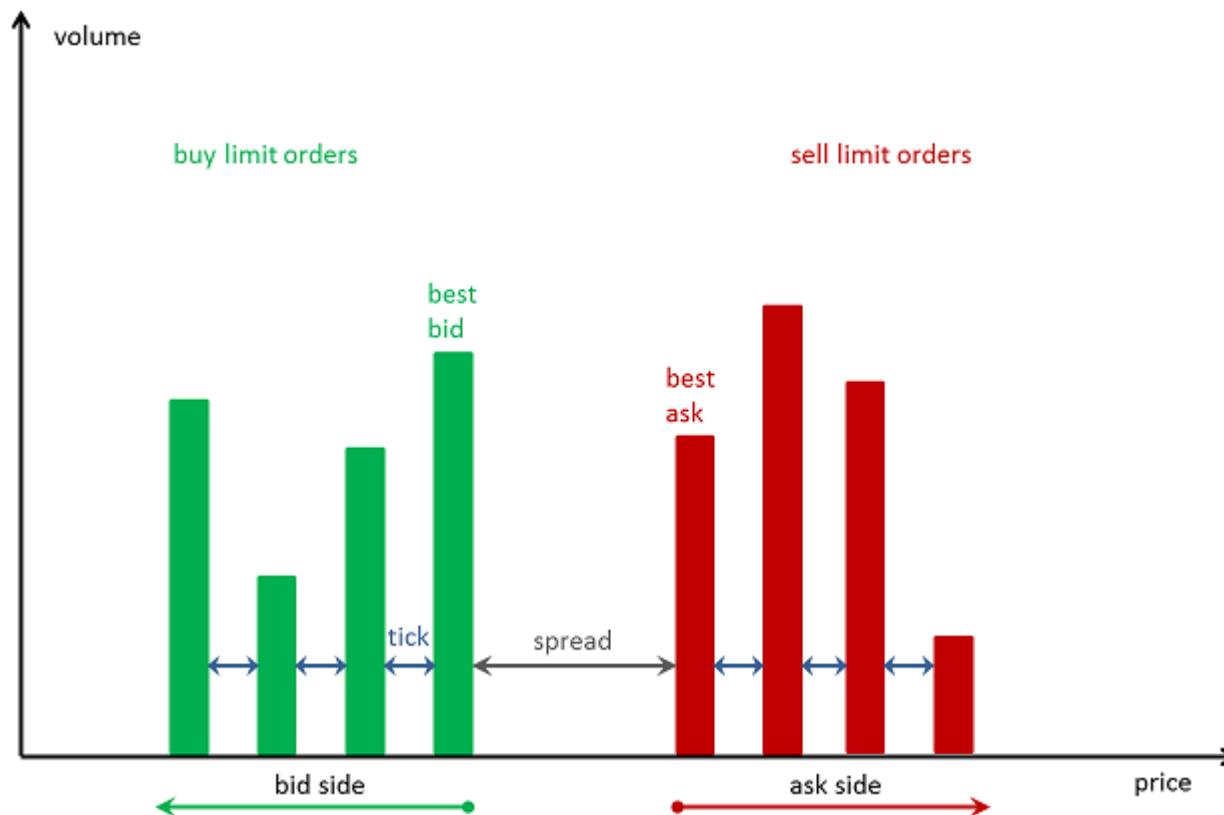
Limit Order Book (LOB)

- If the limit price does not allow a “cross” then the LO will be added to the LOB:
 - If we buy → bid side of the LOB
 - If we sell → ask side of the LOB
- The LOB shows:
 - Available prices
 - Available volumes at each price
 - Number of traders



www.istudiotech.in

LOB: Sketch

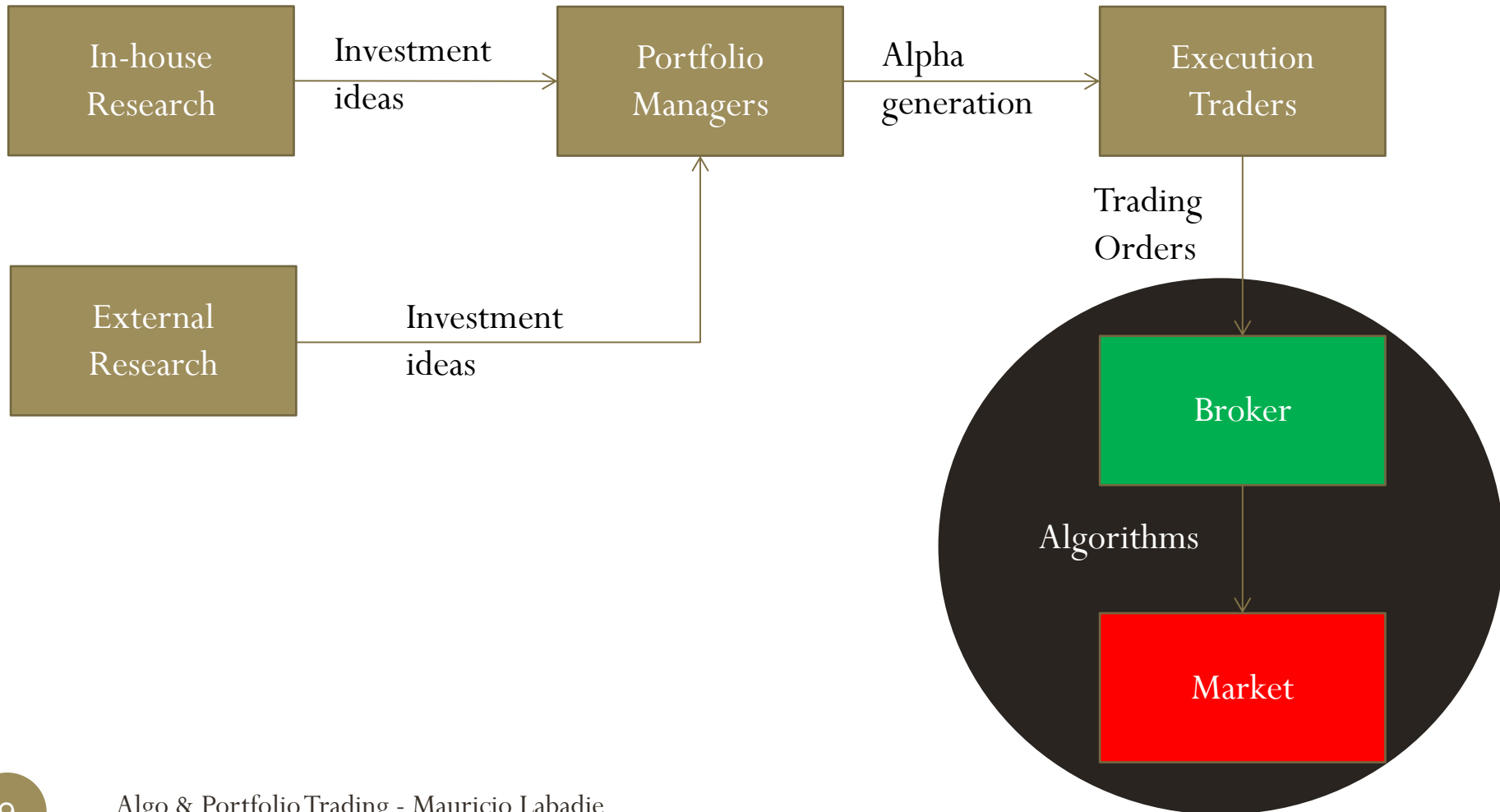


LOB example: LSE

VOD.L VODAFONE GROUP PLC ORD USD0.11 3/7									
Last	AT		162.95					15:42	
Size		2,517	Mid	162.95	Period			SMMP	
Time		15:42	Mid Change	-0.10 (-0.06%)					
Change		-0.10 (-0.06%)	LAT	162.95	Uncross Price		163.25		
Trades		10,457	LAT Change	-0.10 (-0.06%)	Uncross Vol		2,234,586		
Cum Vol		55,950,283	LAT High	163.7	VWAP		162.3293		
LAT Vol		53,810,923	LAT Low	160.75	A-VWAP		162.33515		
P Close		163.05	ISIN	GB00B16GWD56	Market Cap		85,894,857,429		
Open		163.25	NSIN	B16GWD5	P.E.		9.91		
High		163.7	Cur	GBX	Yield		4.82		
Low		160.75	NMS	35,000	Dividend		7.86		
52w Hi		164.4	Segment	SET0	Div-EPS Cur		GBP		
52w Lo		125	Bid Indicator	-	Ex Div		02 Jun 10		
Order Book VOD.L									
4		71006		162.9-162.95		79959		9	
261		6,825,863		155.42523	-167.88226	7,839,197		432	
Cumul	Maker	Size		Bid	Ask	Size	Maker	Cumul	
4		71006		162.90	162.95	79959		9	
10		110436		162.85	163.00	165547		11	
11		194292		162.80	163.05	95435		15	
14		165796		162.75	163.10	246286		18	
16		319872		162.70	163.15	237244		14	
10		224002		162.65	163.20	229145		13	
7		163907		162.60	163.25	304053		13	
4		108296		162.55	163.30	266717		13	
3		90365		162.50	163.35	169815		8	
25		165282		162.45	163.40	177534		7	
1		30702		162.40	163.45	173809		5	

<http://lseg.com>

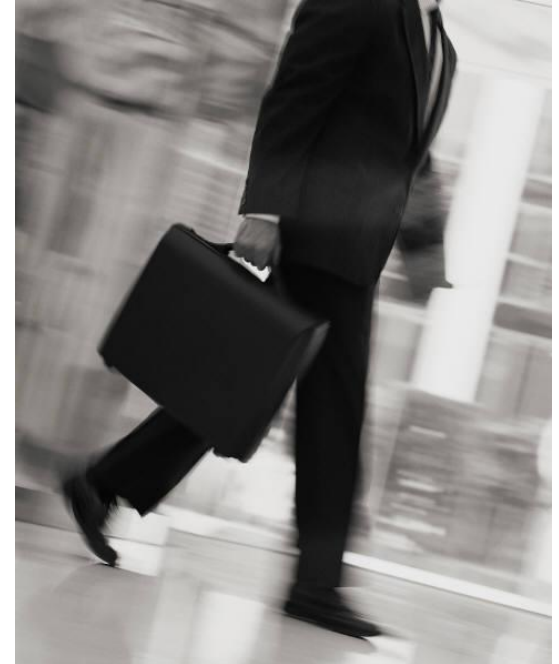
How trading orders get to brokers



2. Portfolio Algorithm

Portfolio algorithm: paradigm

- “The whole is bigger than the sum of the parts”
- If **correlations** are zero then it is the same:
 - To send one portfolio algorithm, or
 - To send N single-name algorithms
- But we know that correlations are not zero:
 - Random variables i.i.d. is a dream
- Using correlations to “fine-tune” a portfolio:
 - Should be more accurate
 - and potentially more profitable



<http://www.paginasprodigy.com/ceocmty>

Portfolio algorithm: natural hedges

- In a portfolio, the risk on some positions can be offset by the risk on others:
 - Long/short portfolios in equities
 - Delta neutral portfolios in options
 - Arbitrage: Indices and Exchange-Traded Funds (ETFs)
- But a portfolio algorithm adds extra complexity:
 - Definition of a portfolio metric
 - Dynamic control of single-name algorithms

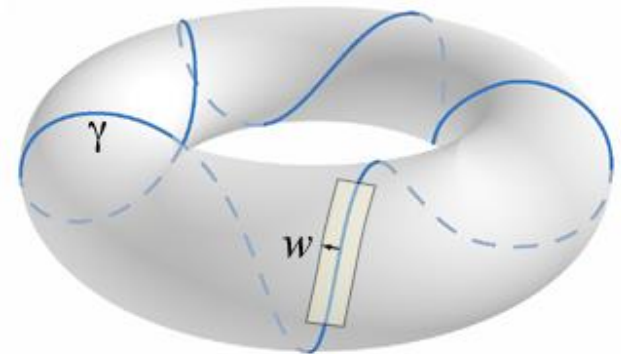


<http://thefinanceassociation.com/>

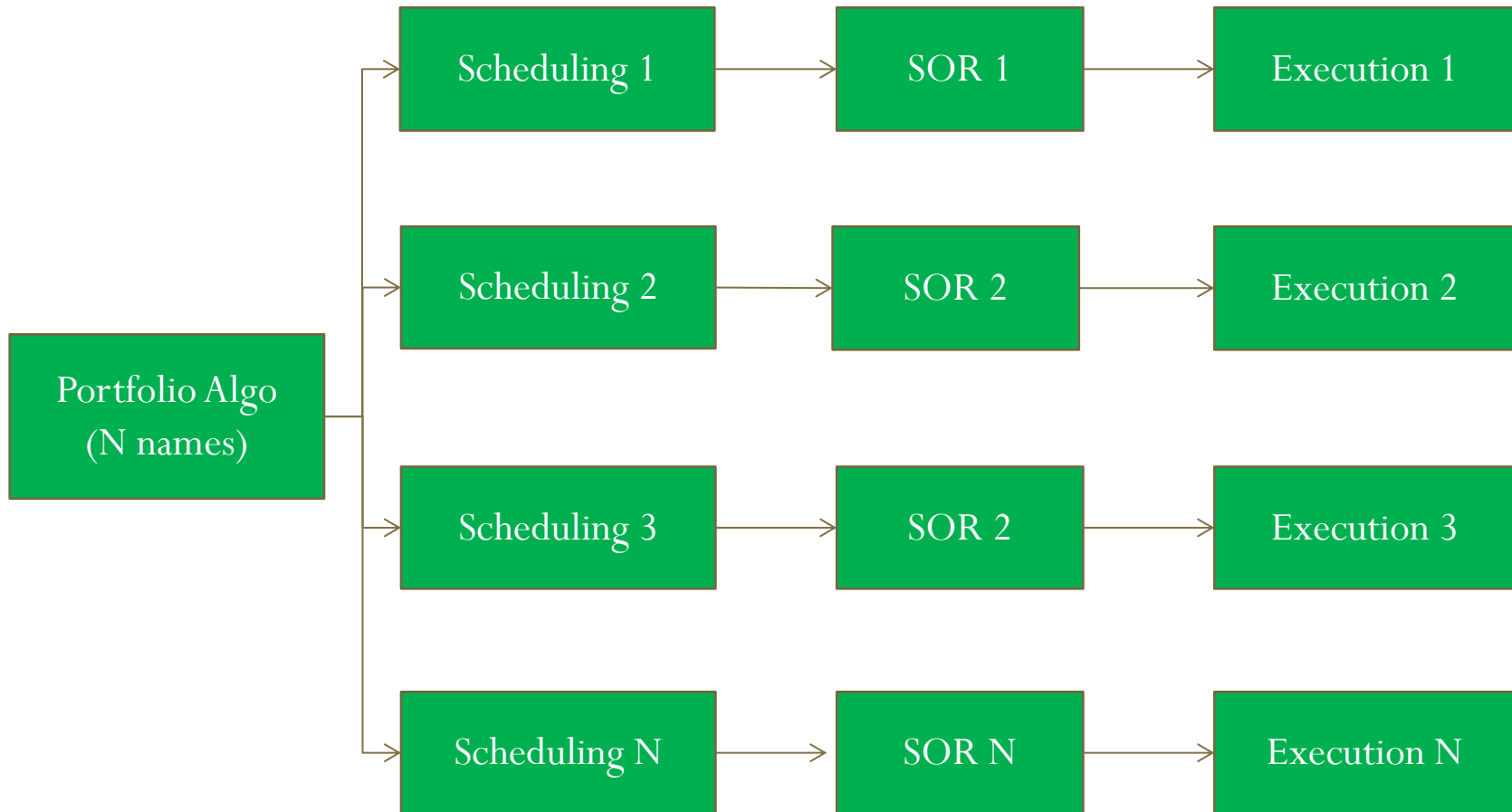
Portfolio algorithm: metric

- This is the constraint at the highest level of hierarchy
 - Single-name algorithms are subordinated to the metric
- It is a quantifiable metric (i.e. a formula) based on:
 - Reward: what is the goal we want to achieve?
 - Risk: volatility, liquidity, news, macro events, etc
- The input can be:
 - Past market conditions: historical
 - Current market conditions: reactive
 - Expected market conditions: forecast
- In a nutshell, the portfolio metric “controls the velocities”:
 - “All single-name algorithms move together”

<http://mathoverflow.net/>



Portfolio algorithm: tree



Portfolio algorithm: metaphor

- Single name algorithms: the horses
- Portfolio metric: the driver
- Risk management: the archer



Source: Florida Center for Instructional Technology, College of Education, University of South Florida.

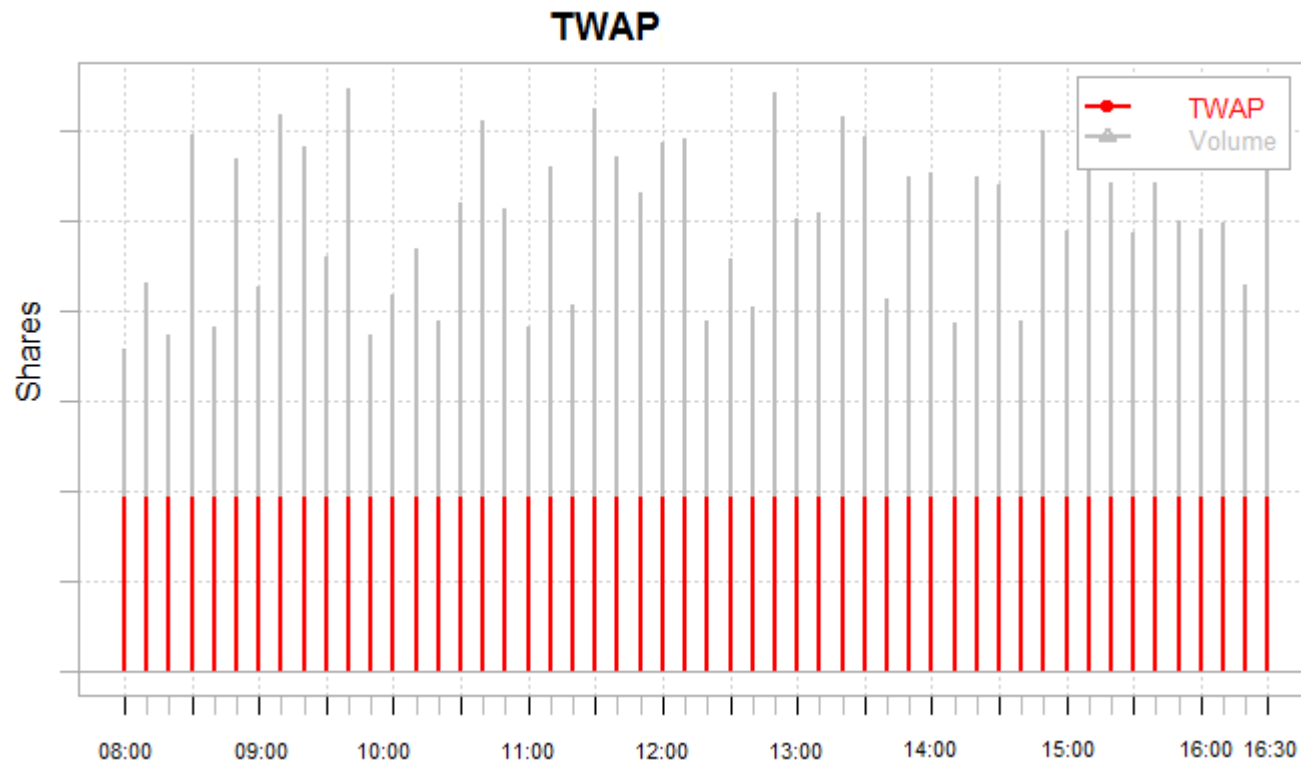
3. Scheduling Algorithms: when to trade?

Scheduling algorithms: TWAP

Time-Weighted Average Price (TWAP)

- It is the simplest scheduling algorithm
- Distributes the total execution evenly in time
- Example:
 - Total number of shares is 85,000
 - Trading hours in London are from 08:00 to 16:30
 - TWAP schedule:
 - 10,000 shares every hour
 - 5,000 shares every 30 minutes
 - 1,667 shares every 10 minutes
 - Etc

Scheduling algorithms: TWAP



Scheduling algorithms: VWAP

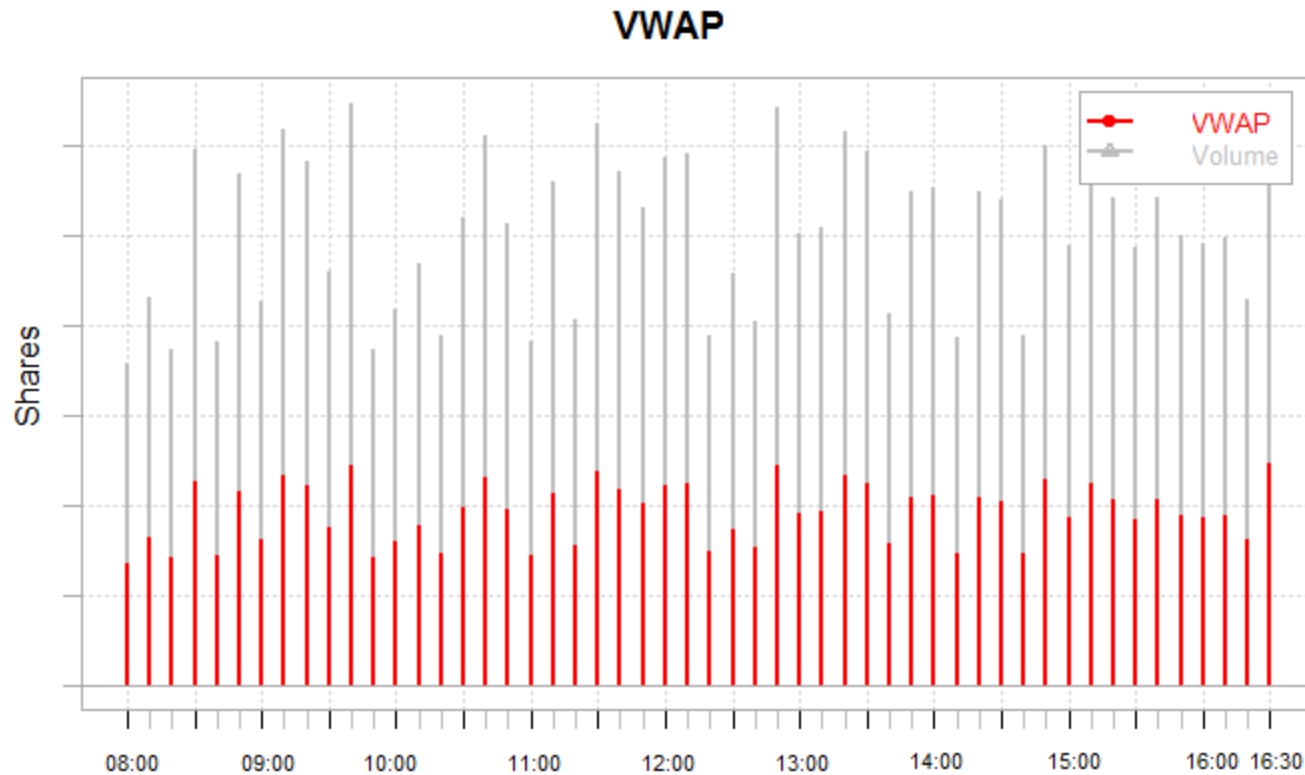
Volume-Weighted Average Price (VWAP)

- The “bread and butter” of scheduling algorithms
- It distributes the total execution according to historical volumes
- It requires a (statistical) volume curve:
 - Divide the day in buckets e.g. 30 minutes
 - For each bucket e.g. from 09:00 to 09:30
 - Compute the average (or median) volume traded between 09:00 and 09:30 in the last 20 days
 - Normalise the volume curve:
 - The sum of all buckets is equal to 1
 - Each bucket represents the percentage of daily volume

Scheduling algorithms: VWAP

- Example:
 - Execute 85,000 shares
 - 09:00 to 09:30 is 4.7% of (daily historical) volume
 - Number of shares: $85,000 * 0.047 = 3,995$
 - 16:00 to 16:30 is 11.2% of volume
 - Number of shares: $85,000 * 0.112 = 9,520$
 - Compare with TWAP schedule:
 - 5,000 shares every 30 minutes
 - 5.9% of volume

Scheduling algorithms: VWAP



4. Smart Order Routing (SOR): where to trade?

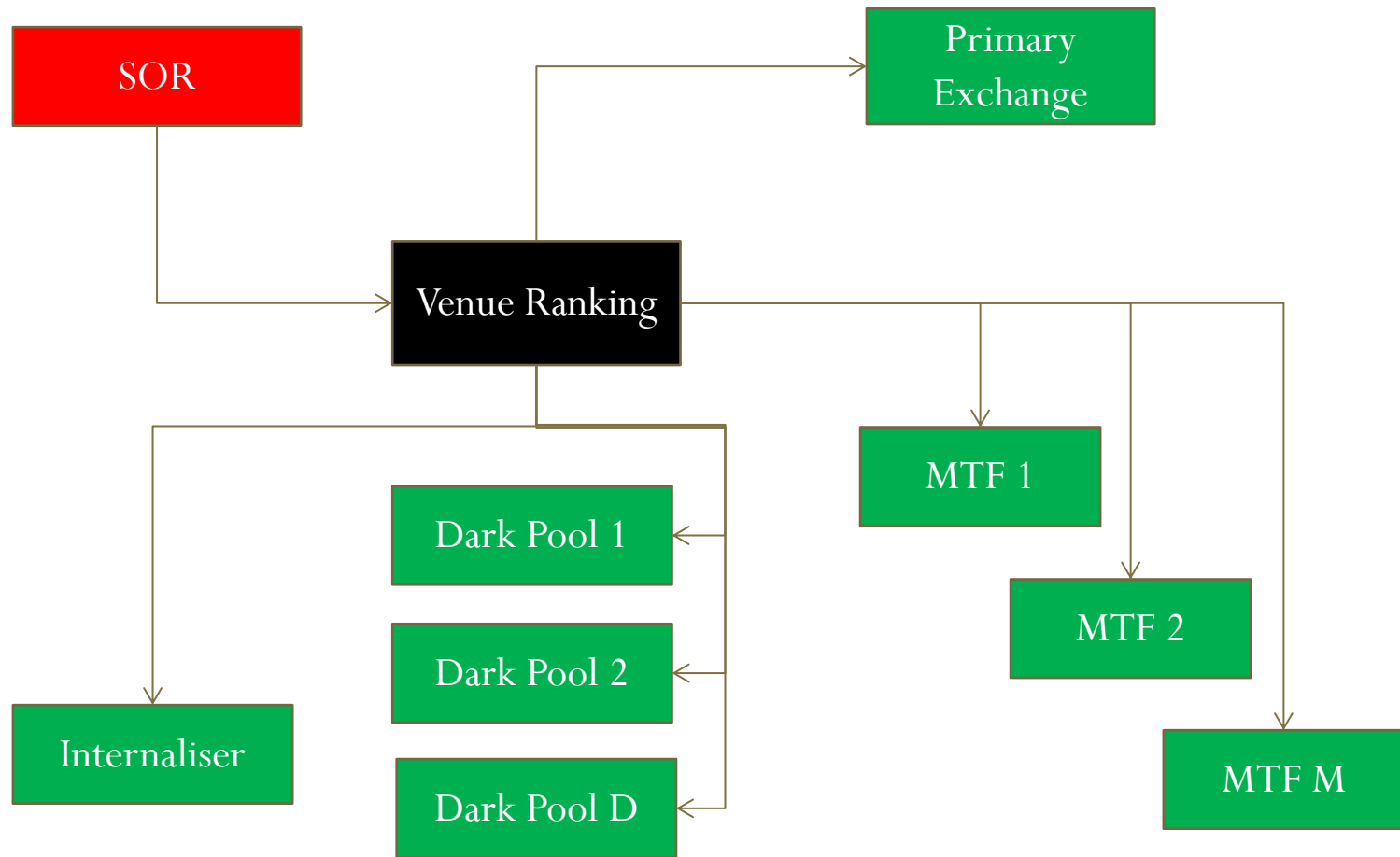
Market Fragmentation

- Primary exchanges:
 - NYSE, NASDAQ, LSE, Euronext, Xetra
- Multilateral Trading Facilities (MTFs):
 - Also known as alternative markets:
 - Chi X, BATS, Turquoise, Aquis
- Dark pools:
 - Block-trading facilities without pre-trade transparency:
 - To reduce market impact
 - But they have to publish post-trade information
- Internalisers:
 - Trade with other desks within the broker/bank

Smart Order Routing (SOR)

- SOR is used to distribute an order into multiple venues:
 - Liquidity seeking in a fragmented market
- Why “smart”?
 - Optimal allocation based on a dynamic “ranking” of venues
- This ranking reflects:
 - Costs e.g. exchange commissions, rebates
 - Volumes e.g. current, historical
 - Signalling e.g. information leakage
 - Indirect costs e.g. spread
 - Quality of liquidity e.g. too many HFTs? Too few?

SOR: Algorithm tree



SOR algorithm: metaphor

- Ranking of venues: the head
- Different venues: the hands



“Thousand-hand Avalokitesvara Bodhisattva” dance

Source: <http://blogs.wsj.com/photojournal/2009/07/01/pictures-of-the-day-210/>.

5. Execution Algorithms: how to trade?

Execution with MOs

- **Fill and Kill / Immediate or Cancel (IoC)**
 - Aggressive Limit order i.e. crossing the spread
 - Tries to execute in one go all shares
 - Non-executed shares are cancelled
 - Possible partial execution
 - Risk of incomplete execution
- **Fill or Kill**
 - IoC with a 100% completion requirement
 - Partial execution is not allowed
 - Risk of no execution

Execution with LOs

- **Peg**
 - Dynamically follow the best limit
 - Always updates to the best bid/ask
 - Always have price priority
 - Try to have as much queue priority as possible
 - Supported by markets
 - Risk of gaming / adverse selection
- **“Float Peg”**
 - Similar to peg order
 - But the “pegged price” can be different to best limit
 - For example, we buy at (best bid – N) ticks
 - Risk of no execution if N is “too large”

Hybrid algorithms

- **Spread Capture**

- Tries to get the best price possible for a passive execution
 - Limit order algorithms in sequence e.g. Peg
- Optimal placing for earning/capturing the spread:
 - “Signals” to determine when and where to post
 - Based on probability of execution

- **Spread Crossing**

- Tries to get the best price possible for an aggressive execution
 - Market order algorithms in sequence e.g. FoK
- Optimal timing for jumping/crossing the spread:
 - Cost of waiting (queue) vs cost of spread
 - Based on probability of execution and “microstructure models”

Hybrid algorithms: metaphor

- Spread Capture: trapper
- Spread Crossing: marksman



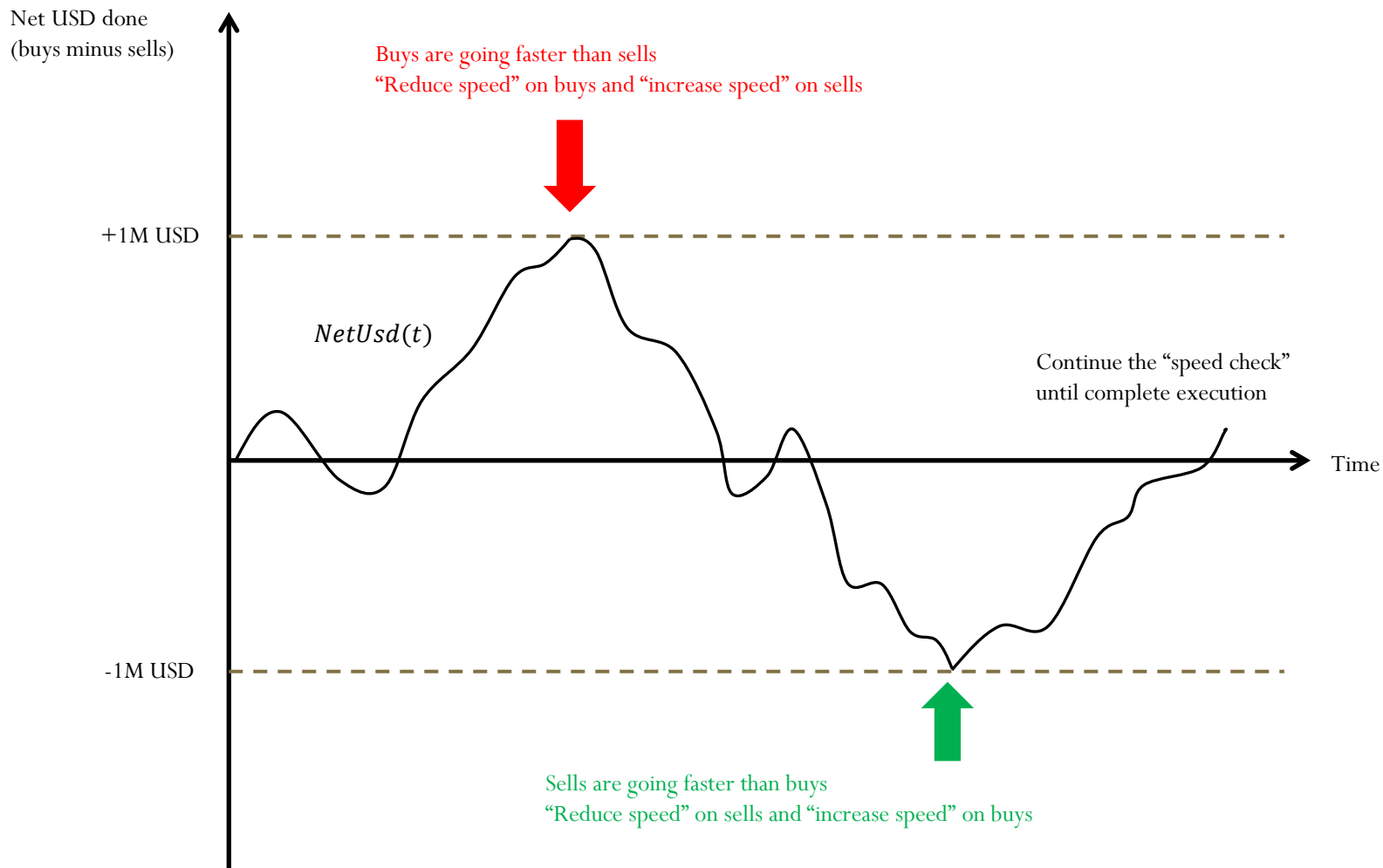
www.deviantart.com & commons.wikimedia.org

6. Example of Portfolio algorithm

Turnover of portfolio

- Turnover of position from Telecoms to Pharma:
 - Sell 100M USD worth of Telecoms
 - Buy 100M USD worth of Parma
 - Proceedings from the sells will finance the buys
- Scheduling algorithm:
 - TWAP
- Market Neutral metric:
 - $Metric(t) = |ValueBuysUsd(t) - ValueSellsUsd(t)|$
 - It “naturally” reduces market risk
 - If the market goes up we lose on our buys
 - But we win on our sells

Turnover of portfolio



Portfolio algorithm

- White horse: scheduling algorithm (TWAP tracker)
- Black horse: metric (Market Neutral)
- Driver: cost function (or metric)
- Archer: monitoring of real time market data



Source: Florida Center for Instructional Technology, College of Education, University of South Florida.

References

- Harris, “*Trading and Exchanges*”, Oxford University Press, 2003
- Barry Johnson, “*Algorithmic Trading & DMA*”, 4Myeloma Press, 2010
- Labadie, “*Optimal algorithmic trading and market microstructure*”. MSc thesis
- Labadie, Lehalle “*Optimal starting times, stopping times and risk measures for algorithmic trading*”, Journal of Investment Strategies (Risk) Volume 3, Number 2 (December 2014)
- Cartea, Jaimungal, Penalva, “*Algorithmic and High Frequency Trading*”, Cambridge University Press, 2015



LOTR Movie

Thank you for your
attention

